

Laboratory Implementation of Direct Torque Controller based Speed Loop Pseudo Derivative Feedforward Controller for PMSM Drive

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Abstract—This paper, evaluate the effectiveness of a proposed speed loop pseudo derivative feedforward (PDFF) controller-based direct torque controller (DTC) for a PMSM drive against the performance of existing PI speed controller-based DTC and hysteresis current controller (HCC). The proposed PDFF-based speed regulator effectively reduces oscillation and overshoot associated with rotor angular speed, electromagnetic torque, and stator current. Two case studies, one using forward-to-reverse motoring operation and the other involving reverse-to-forward braking operation, has been validated to show the effectiveness of the proposed control strategy. The proposed controller's superior performance is demonstrated through experimental verification utilizing an FPGA controller for a 1.5 kW PMSM drive laboratory prototype.

Index Terms—Direct torque control, Pseudo derivative feedforward controller, Permanent magnet synchronous motor (PMSM).

I. INTRODUCTION

RESENT trends in the industry and research, PMSM has gained significant importance due to high-power density, power factor and its efficiency. In general, hysteresis current control [1], [2] is used to provide sinusoidal reference currents to the motor, resulting in high dynamic performance of the PMSM drive. Direct Torque Control (DTC) has grown in popularity because it has less parameter dependency and eliminates the need for coordinate transformation [3]-[12]. Due to its advantages DTC is widely implemented in control of different machines

viz., synchronous reluctance machine, permanent magnet synchronous motor and induction motor. Current regulators and pulse with modulation (PWM) are not used in the DTC scheme [4]-[6]. In case of PMSM drives based on DTC the main focus to obtain fast transient response of electromagnetic torque and stator current ripples compared to Field Oriented Control (FOC) [9]. Further, DTC offers superior torque performance in both steady-state and transient scenarios. The electromagnetic torque and flux are the main quantities involved in DTC. The speed loop controller that produces the reference electromagnetic torque is input to the DTC. Further flux and torque estimator [12] in DTC structure generates the feedback signal. The implementation of PI controller-based speed control of PMSM drives has an issue with overshoot and settling time. In this regard to increase the settling time the gains of the PI controller should be increased which further leads to oscillations and overshoot to the system. Further the aforementioned issues arises due to presence of load disturbances in the system also. Therefore, sliding mode control is proposed in [13]-[15] to improve the drive performance under uncertainties in machine parameters along with load disturbance. Still the chattering phenomenon in sliding mode control creates disturbances in the system. In [16]-[20], due to its dependence on machine parameters, the DTC method based on the duty cycle control technique suffers from uncertainties. In recent years, many researchers focused on addressing challenges associated with switching table and duty cycle control [3]-[12], [16]-[23] to reduce ripples in electromagnetic torque and stator current at steady state condition. In [24]-[29], the PMSM is controlled through cascaded PI-PI controller, Fuzzy-PI Controller, Model Predictive Speed Controller, extended Kalman filter based speed control, nonlinear sliding mode speed control, genetic algorithm based PI speed controller is implemented. These control schemes require rotor position information to control the PMSM drive. But in this paper the main focus is to enhance the performance of PMSM drive during transient condition by adopting PDFF controller-based speed regulator with existing switching table-based DTC. The proposed speed regulator significantly reduces overshoot in the speed, electromagnetic torque and stator current. In [30], simulation results are carried out for four quadrant operation

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of PMSM drive. In this paper, experimental results for four quadrant operation of PMSM is presented to prove the robustness of the proposed controller. Finally, comparative analysis based on experimental studies are carried out for PI-HCC, PI-DTC and PDFF-DTC schemes.

The performance of the PMSM drive with the speed loop pseudo derivative feedforward controller is discussed in this study. To demonstrate the robustness of the proposed controller, a comparison with PI-based DTC and HCC is also shown. The paper is structured as follows: Section II, describes a mathematical model of the PMSM drive. In Section III, PI speed controller-based DTC scheme and the PI speed controller based HCC system, as well as the PDFF speed controller based DTC scheme (i.e., structure, design procedure and time response analysis) is discussed. Finally, experimental results are demonstrated and comparative study is consolidated in the section IV.

II. MODELLING OF PMSM DRIVE IN ROTOR REFERENCE FRAME

The dynamic model of PMSM is modelled in d - q axes reference frame. The stator voltages in d - q axes can be described as [21]:

$$v_{qs} = R_s i_{qs} + \omega_r \psi_{ds} + p \psi_{qs} \quad (1)$$

$$v_{ds} = R_s i_{ds} - \omega_r \psi_{qs} + p \psi_{ds} \quad (2)$$

The stator flux in d - q axes can be expressed as:

$$\psi_{ds} = L_s i_{ds} + \psi_m \quad (3)$$

$$\psi_{qs} = L_s i_{qs} \quad (4)$$

Substitution of (3)-(4) in (1)-(2) respectively, current in stator is depicted as

$$\frac{d}{dt} \begin{pmatrix} i_{ds} \\ i_{qs} \end{pmatrix} = \begin{pmatrix} -\frac{R_s}{L_s} & \omega_r \\ -\omega_r & -\frac{R_s}{L_s} \end{pmatrix} \begin{pmatrix} i_{ds} \\ i_{qs} \end{pmatrix} + \begin{pmatrix} \frac{1}{L_s} & 0 \\ 0 & \frac{1}{L_s} \end{pmatrix} \begin{pmatrix} v_{ds} \\ v_{qs} \end{pmatrix} + \begin{pmatrix} 0 \\ -\frac{\psi_m}{L_s} \end{pmatrix} \omega_r \quad (5)$$

The computation of rotor speed, position and electromagnetic torque are expressed as follows:

$$T_e = \frac{3}{2} \cdot \frac{P}{2} (\psi_{ds} i_{qs} - \psi_{qs} i_{ds}) \quad (6)$$

$$T_e - T_l = J \frac{d\omega_r}{dt} \quad (7)$$

$$\theta_r = \int \omega_r dt \quad (8)$$

III. SYSTEM DESCRIPTION

A. DTC with Speed Loop PDFF Controller

The schematic of a PMSM drive using a DTC-based speed loop PDFF controller is shown in Fig. 1. This DTC scheme is implemented in stator flux frame, which requires knowledge of stator flux in flux-torque (M-T) axes. The torque axis involves dual loops: outer loop speed control and inner loop torque control. On other hand, stator flux axis consists of single loop. Utilizing a PDFF controller, the outer speed loop creates an electromagnetic torque reference for torque control.

In [9], overshoot in rotor speed is discussed for a DTC-based PMSM drive with speed loop PI control, overshoot is unavoidable during system dynamics perturbation. The overshoot and speed oscillations are reduced by the proposed PDFF speed regulator. An error signal of electromagnetic torque and stator flux is generated by comparing measured and estimated values of these quantities. Then, the DTC's switching table is used to generate logic for switching based on the error signals processed through hysteresis comparator. In Table I, a switching table for DTC [9] is shown.

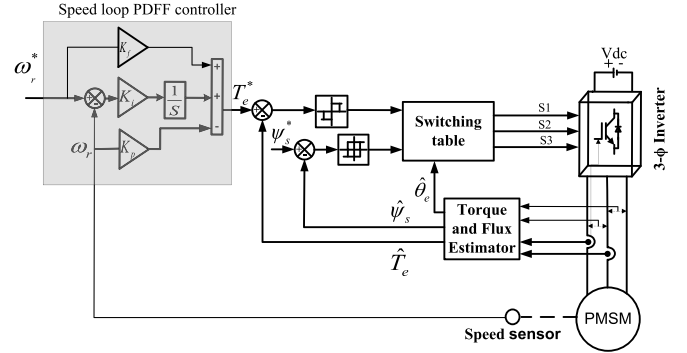


Fig. 1. Proposed control of PMSM drive.

TABLE I
DTC BASED SWITCHING

ψ	T	θ					
		θ_1	θ_2	θ_3	θ_4	θ_5	θ_6
-1	-1	001	101	100	110	010	011
	0	000	111	000	111	000	111
	1	010	011	001	101	100	110
1	-1	101	100	110	010	011	001
	0	111	000	111	000	111	000
	1	110	010	011	001	101	100

The stator fluxes are computed using (9) and (10) in the stationary reference frame (α - β). The resultant stator flux is computed using (11).

In the α - β reference frame, estimated stator flux is expressed as:

$$\hat{\psi}_{\alpha s} = \int (v_{\alpha s} - R_s i_{\alpha s}) dt \quad (9)$$

$$\hat{\psi}_{\beta s} = \int (v_{\beta s} - R_s i_{\beta s}) dt \quad (10)$$

where $\hat{\psi}_{\alpha s}$ and $\hat{\psi}_{\beta s}$ are the estimated stator flux.

$$\hat{\psi}_s = \sqrt{(\hat{\psi}_{\alpha s}^2 + \hat{\psi}_{\beta s}^2)} \quad (11)$$

$$\hat{\theta}_s = \sin^{-1} \left(\frac{\hat{\psi}_{\beta s}}{\hat{\psi}_s} \right) \quad (12)$$

Estimated electromagnetic torque is described as:

$$\hat{T}_e = \frac{3}{2} \cdot \frac{P}{2} (\hat{\psi}_{\alpha s} i_{\beta s} - \hat{\psi}_{\beta s} i_{\alpha s}) \quad (13)$$

B. DTC with PI Speed Loop Controller

The PI-DTC strategy [2] consists of torque and flux axis. Torque axis consists of dual loops: speed control in outer loop (PI) and torque control in inner loop. On other hand, flux axis consists of single loop. Consider the outer loop, which compares the reference and actual rotor speeds to

generate speed error, which is then fed into the PI controller of the speed loop to generate the torque controller reference. In order to produce the reference for the torque controller, the reference speed and the actual rotor speed are compared to produce a speed error. In the stator flux control, rated value of stator flux is considered as reference. The error signal of electromagnetic torque and stator flux is produced by comparing reference electromagnetic torque and stator flux with the estimated electromagnetic torque and stator flux respectively. Then, DTC will take inputs of instantaneous electromagnetic torque and stator flux error signals and processed to generate the switching pulse for 2-level inverter.

C. HCC with PI Speed Loop Controller

The PI-based HCC (FOC) [1] approach is considered because of its easy implementation. This scheme is implemented in rotor reference (d - q axes) which comprise PI control for outer speed loop and inner loop with HCC. The q -axis consists of two loop which represented as outer speed loop PI control and HCC as inner loop. The reference q -axis stator current is produced by the PI-based speed controller. On the other hand, d -axis involve single loop i.e., inner loop HCC where the reference zero is considered as reference for d -axis stator current. Then, three phase reference currents (i_a^* , i_b^* , i_c^*) are obtained based on d - q axes reference stator currents using inverse park transformation. Further, the instantaneous stator current error is calculated by comparing the reference and measured currents. This error signals processed into the hysteresis band, which eventually produces firing pulses for the VSI.

D. Effectiveness of Proposed Speed Loop PDFF and PI Controller

In this section, outer speed loop is examined to evaluate the effectiveness of PI-HCC, PI-DTC, and the proposed PDFF-DTC scheme. The simulation results show that the dynamic performance of the proposed system (overshoot and settling time) is superior then the PI controller-based speed regulator. Fig. 2 depicts the concept of a PDFF controller-based speed regulator for a DTC-PMSM drive. In PI-HCC and PI-DTC based PMSM drives, the outer speed loop is comprise of a PI controller. Thereby in the analysis, the same PI controller gain values are considered for both PI-HCC and PI-DTC based PMSM drive, and the structure of the outer speed loop is shown in Fig. 3. By experimenting with different controller gain values, as shown in Fig. 4, the efficacy of outer loop PI and PDFF controller-based speed regulator is validated. The results are pertained to step response of speed regulator at no load condition i.e., load torque disturbance is considered as zero. It is clear that the PDFF controller overshoots the speed by 7.65% to 0, whereas the PI controller overshoots it by 10.65%. It's clear that, the PDFF controller-based system is more stable and has less overrun than the PI controller-based system. Bode plot of the PDFF and PI speed controller is shown in Fig. 5. Overshoot is mitigated by the PDFF controller for the speed

loop, which has a significantly smaller magnitude gain than the PI-based speed loop system.

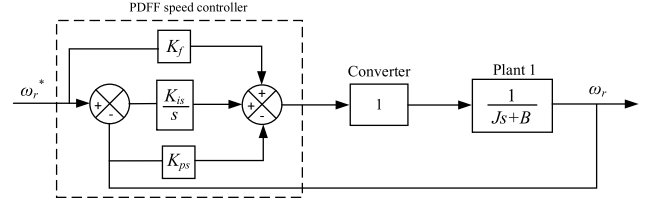


Fig. 2. Speed loop PDFF controller structure.

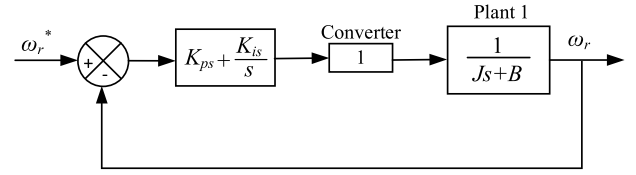


Fig. 3. PI controller-based speed loop structure.

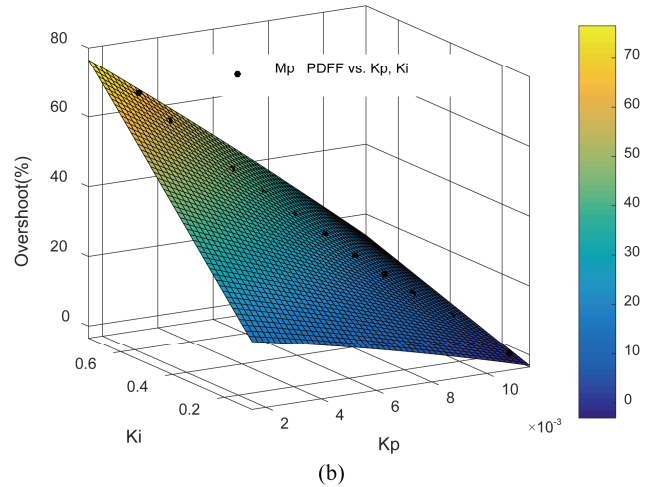
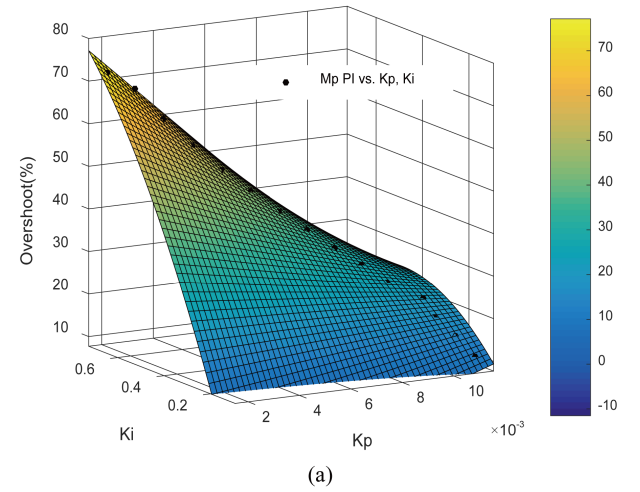


Fig. 4. Overshoot for various values of K_i and K_p . (a) PI controller. (b) PDFF controller.

E. Design of PI and PDFF Speed Controller

In this paper, extended root locus technique (ELRT) is used to design the controller parameters. Table II shows the closed loop transfer function of a PI speed controller. Fig. 6(a) shows

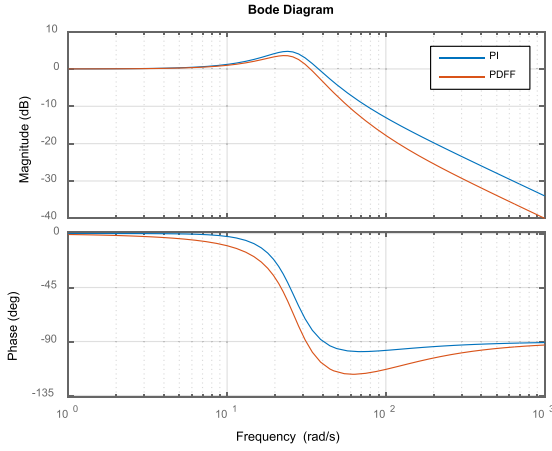


Fig. 5. Bode plot response of PI and PDFF controller of speed regulator. the ERLT for tuning the speed loop PI controller. To determine optimal value of K_{ps} , root locus is plotted for different values of K_{is} ($0 \leq K_{is} \leq 0.6$, with interval of 0.1). The proposed PDFF speed controller is simple and non-complex, with no high computational complexity or chattering. Table II shows the closed loop transfer function of a speed loop PDFF controller. The PDFF speed controller consists of three gains, i.e., K_{ps} , K_{is} and K_f where K_f is αK_{ps} and α range is between 0 to 1 and its chosen as 0.4 in this paper. The K_{ps} and K_{is} values are determined for the PDFF controller design using extended root locus method. The set of K_{is} ($0 \leq K_{is} \leq 0.6$, with interval of 0.1) values are considered and plotted set of root locus of K_{ps} and monitored the bandwidth root locus is shown in Fig. 6(b). The K_{is} value keep on increased and plotted root locus of K_{ps} . For the value of K_{is} , root locus of K_{ps} reaches the closed loop bandwidth and is considered as optimal value of K_{is} . Further, optimal value of K_{ps} is computed from the root locus curve for

TABLE II
TRANSFER FUNCTION FOR PI AND PDFF SPEED CONTROLLER

PI based speed controller	PDFF based speed controller
$\frac{\omega_r}{\omega_r^*} = \frac{sK_{ps} + K_{is}}{s^2J + s(B + K_{ps}) + K_{is}}$	$\frac{\omega_r}{\omega_r^*} = \frac{sK_f + K_{is}}{s^2J + s(B + K_{ps}) + K_{is}}$

where $K_f = \alpha K_{ps}$, $0 \leq \alpha \leq 1$

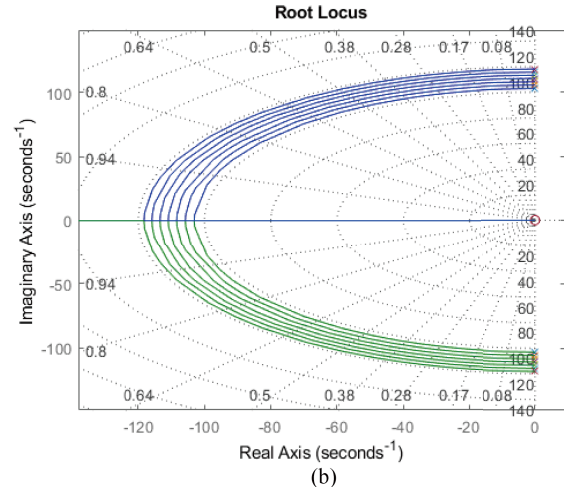
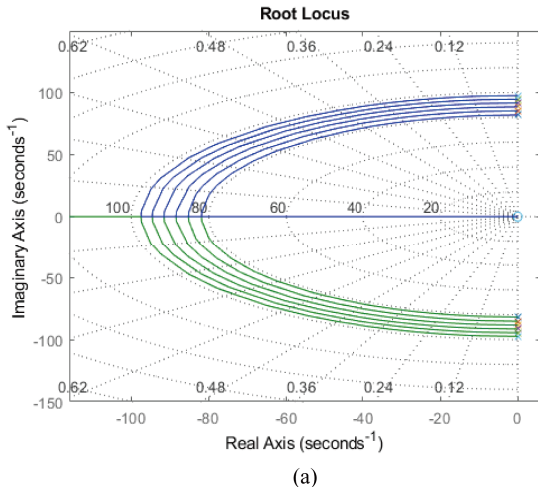


Fig. 6. Tuning procedure for speed controller. (a) Calculating K_{ps} for PI speed controller. (b) Calculating K_{ps} for PDFF speed controller.

TABLE III
STABILITY ANALYSIS OF PI AND PDFF SPEED CONTROLLER

Parameters	PI controller	PDFF controller
Gains	K_{ps} and K_{is}	K_{ps} , K_{is} and K_f
Relative Stability	97.8	118
Damping Factor (ζ)	1	1

optimal value of K_{is} at $\zeta=1$. Table III shows the relative stability for PI and PDFF speed controller, it is inferred that relative stability of the proposed PDFF speed controller is improved than PI speed controller for $\zeta=1$.

F. Time Response Analysis for PI and PDFF Speed Controller

The step change in the speed response of the PDFF, the PI controller, and the feedforward gain variation of the PDFF controller are discussed in detail in this section. Fig. 7 shows a comparison of the step change in speed response of the PDFF and PI controllers for the PMSM drive. The PI controller's speed response has a maximum overshoot of 30.25% at $\zeta=0.5$ and a minimum overshoot of 13.7% at $\zeta=1$. The PDFF controller's speed response has a maximum overshoot of 19.2% at $\zeta=0.5$ and eliminates overshoot at $\zeta=1$. For better performance ζ is chosen as 0.7. It can be seen from Fig. 7, the PI controller has on overshoot of 21.4% with a settling time of 85 ms ($\zeta=0.7$), while the PDFF controller-based system has 6.8% overshoot with a settling time of 50 ms ($\zeta=0.5$). As a result, the PDFF controller improves the dynamic response of the PMSM drive (i.e., improves the settling time and reduces the overshoot than that of PI controller based PMSM drive).

The performance of PDFF speed controller is analyzed for feedforward gain variation as shown in Fig. 8. The feedforward gain K_f is defined as αK_{ps} where α range is between 0 to 1. When α is equal to zero, the PDFF controller will act as PDF controller. On the hand if α is equal to one, then the PDFF controller will act as PI controller. Thus, PDFF controller possess inherit feature of exhibiting of PI and PDF characteristics based on the α value. For the better effectiveness i.e., quick rise time and less overshoot when K_f value is chosen as $0.4K_{ps}$.

Fig. 9 shows the performance of both PI-DTC and PDFF-DTC based PMSM drive. It is concluded that the PI-DTC based system exhibits an overshoot of 17.8% during speed variation from standstill to the rated (75% of the full load). On the other hand, the proposed PDFF-DTC based system exhibits a maximum overshoot of 3% for the aforementioned condition which is very less compared to PI-DTC based system.

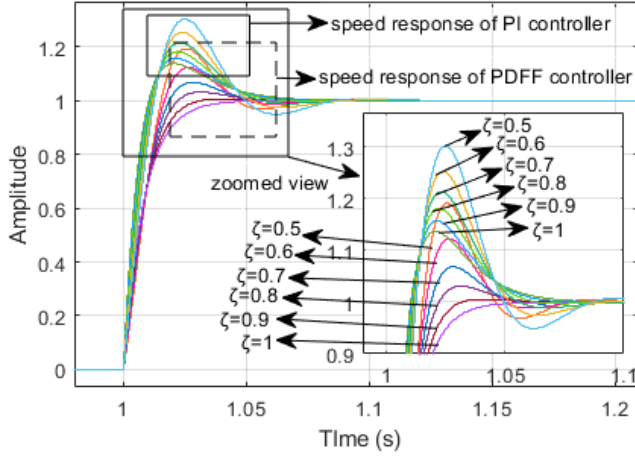


Fig. 7. Comparison of PDFF and PI controller under damping coefficient (ζ) variation.

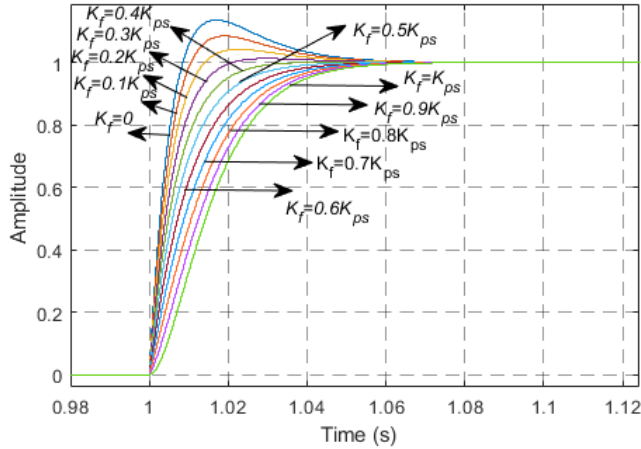


Fig. 8. Performance of PDFF controller during variation in the feedforward gain (K_f).

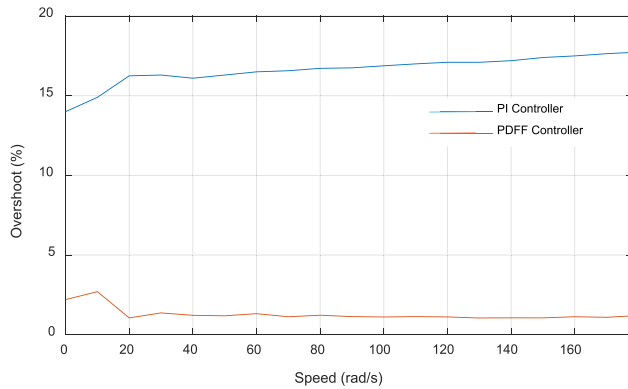


Fig. 9. The overshoot vs rotor speed, time response of the proposed PDFF-DTC and conventional PI-DTC system.

IV. EXPERIMENTAL RESULTS

This section pertains to demonstration of experimental results for four quadrant operation for a PMSM drive of 1.5 kW. Experimental implementation for the proposed scheme is realized with FPGA controller (ALTERA cyclone II). The signal conditioning and sensing circuitry is built up of Op-Amps with LEM current and voltage sensors using OPA 227P, LAH 25-NP, LV-25P respectively. Conclusively, a 1024 PPR encoder is employed for gathering rotor angular speed and rotor position of motor. The experimental setup is depicted in a block diagram (Fig. 10) and the laboratory prototype is illustrated in Fig. 11. The comparative analysis of PMSM drive is performed for speed loop PI controller based hysteresis current controller (HCC), speed loop PDFF and conventional PI controller based DTC under different case studies viz. case (1): reverse to forward braking operation and case (2): forward to reverse motoring operation.

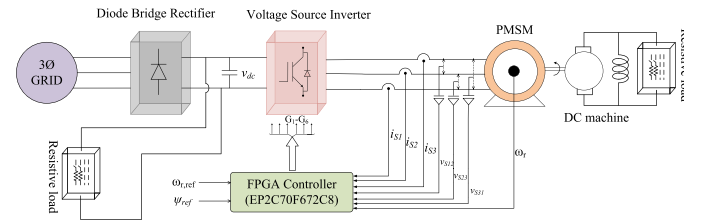


Fig. 10. Block diagram of experimental setup.

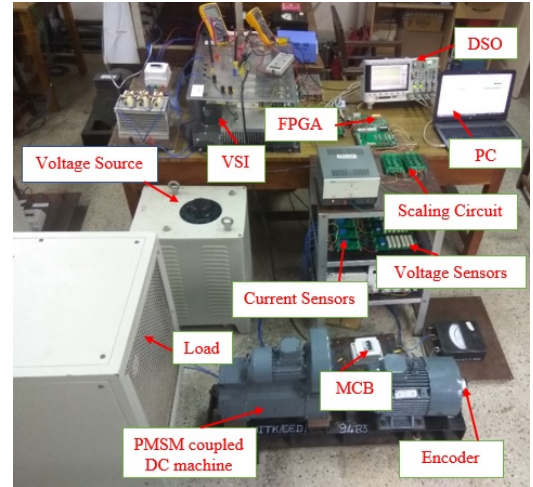


Fig. 11. Photograph of laboratory setup.

Case (1): Reverse to Forward Braking Operation

Fig. 12 demonstrates the experimental results for reverse to forward braking operation. To validate the system performance, variation is considered in the speed (from -157.2 rad/s to 157.2 rad/s) and load (from -8.2 N · m to -8.0 N · m). The speed response of the PI-HCC and PI-DTC systems are depicted in Fig. 12(a) and (b) respectively. The PI-HCC system have a overshoot of 140% and a steady state time of 85 ms in rotor speed, while the rotor speed has overshoot of 130% and a steady state time of 75 ms with PI-DTC system. Fig. 12(c) depicts that, the rotor speed has an overshoot of 56% with settling time of 55 ms for the proposed PDFF-DTC system. So, it is found that with PDFF-DTC system the over-

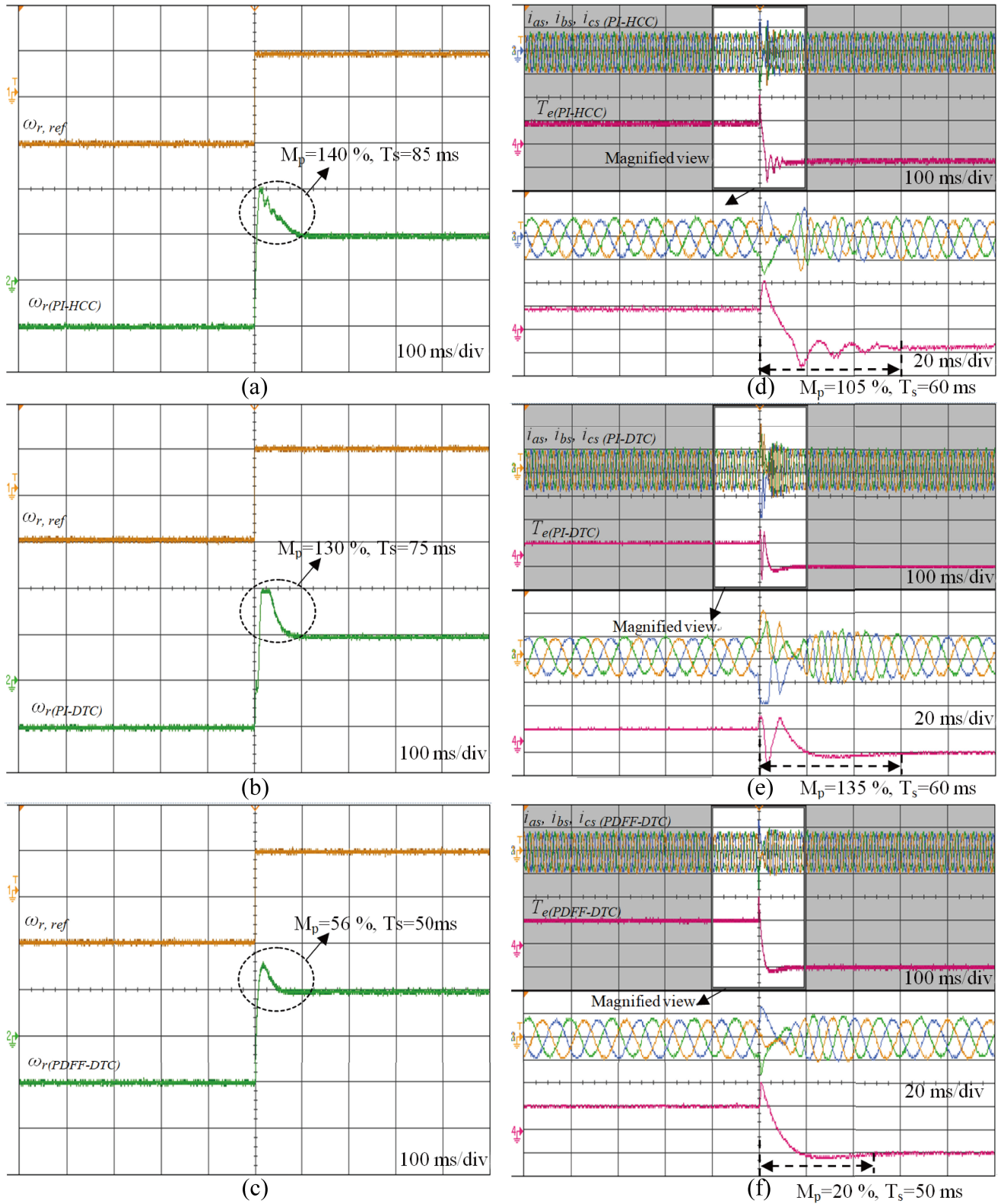


Fig. 12. Reverse to Forward braking operation. (a) Reference, rotor speed (PI-HCC), Scale: ($\omega_{r(ref)}$, $\omega_{r(aut)}$) = 160 rad/s/div). (b) Reference, rotor speed (PI-DTC), Scale: ($\omega_{r(ref)}$, $\omega_{r(aut)}$) = 160 rad/s/div). (c) Reference, rotor speed (PDFF-DTC). Scale: ($\omega_{r(ref)}$, $\omega_{r(aut)}$) = 160 rad/s/div). (d) Electromagnetic torque, Three phase stator current (PI-HCC), Scale: (T_e = 10 Nm/div), (i_{as} , i_{bs} , i_{cs}) = 8 A/div). (e) Electromagnetic torque, stator three phase current (PI-DTC), Scale: (T_e = 4 Nm/div), (i_{as} , i_{bs} , i_{cs}) = 8 A/div). (f) Electromagnetic torque, stator three phase current (PDFF-DTC), Scale: (i_{as} , i_{bs} , i_{cs}) = 8 A/div), (T_e = 10 Nm/div).

shoot in rotor speed has been reduced by 55% compared to PI-HCC & PI-DTC system. Fig. 12(d)-(f) shows the electromagnetic torque and stator three phase current response. It is clear that, with the proposed controller the

electromagnetic torque and stator current has smooth transient response compared to other controllers i.e. PI-HCC and PI-DTC. With the PI-DTC, PI-HCC and PDFF-DTC system, the response of electromagnetic torque has an

overshoot of 105%, 135% and 20% respectively with the settling time of 60 ms, 60 ms & 50 ms respectively. On the other hand the stator current response has an overshoot of 80%, 215% & 70% respectively with settling time of 60 ms, 60 ms and 50 ms respectively.

Case (2): forward to Reverse Motoring Operation

Fig. 13 demonstrates the experimental results for forward to reverse motoring operation. In this scenario, simultaneous variation of load (8 N·m to -8 N·m) and speed (157 rad/s to -157 rad/s) is considered. Forward to reverse motoring operati-

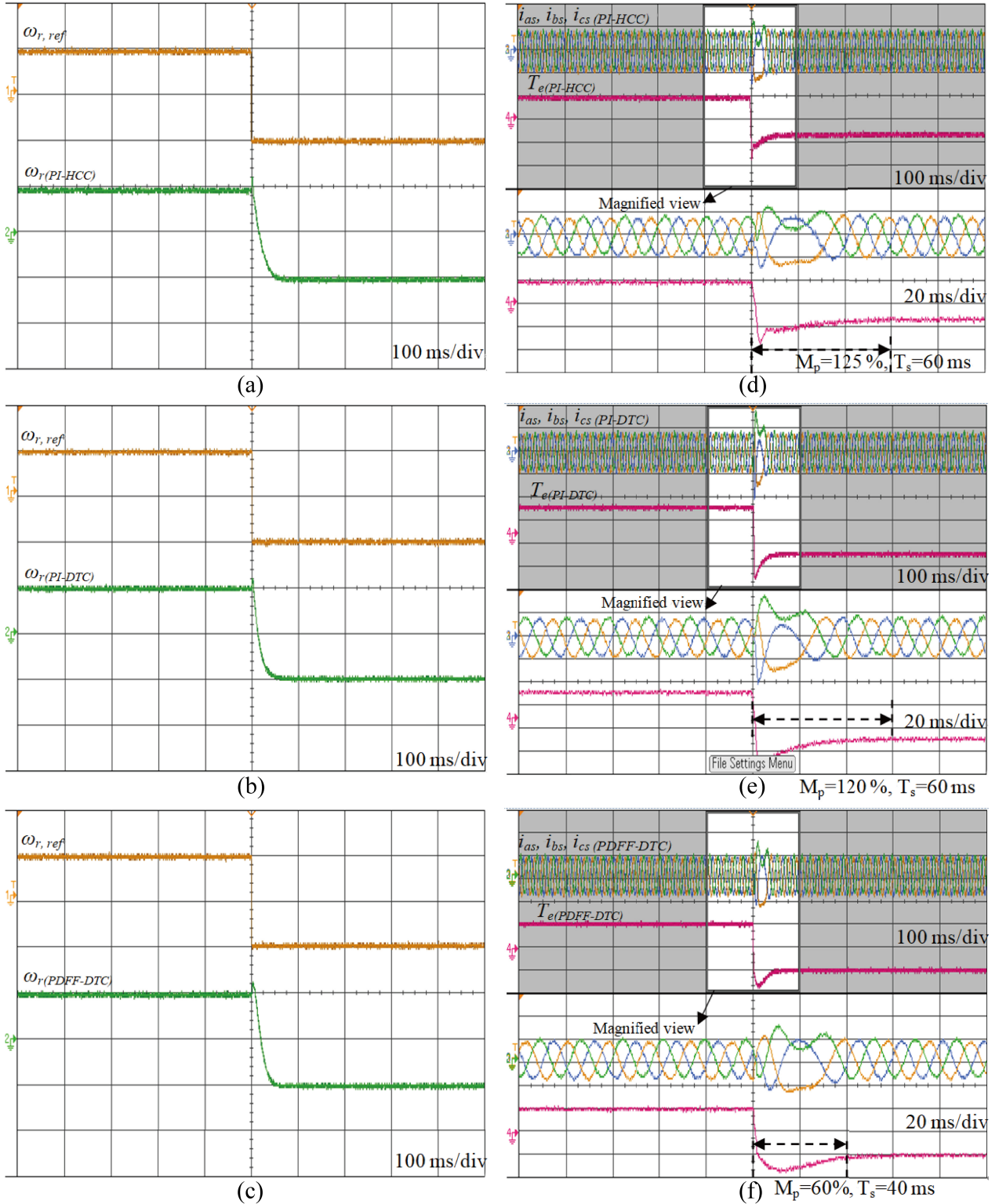


Fig. 13. Forward to Reverse motoring operation. (a) Reference, rotor speed (PI-HCC), Scale: ($\omega_{r(ref)}$, $\omega_{r(aut)}$) = 160 rad/s/div). (b) Reference, rotor speed (PI-DTC), Scale: ($\omega_{r(ref)}$, $\omega_{r(aut)}$) = 160 rad/s/div). (c) Reference, rotor speed (PDFF-DTC). Scale: ($\omega_{r(ref)}$, $\omega_{r(aut)}$) = 160 rad/s/div). (d) Electromagnetic torque, stator three phase current (PI-HCC), Scale: (T_e = 10 Nm/div), (i_{as} , i_{bs} , i_{cs} = 8 A/div). (e) Electromagnetic torque, stator three phase current (PI-DTC), Scale: (T_e = 4 Nm/div), (i_{as} , i_{bs} , i_{cs} = 8 A/div). (f) Electromagnetic torque, stator three phase current (PDFF-DTC), Scale: (i_{as} , i_{bs} , i_{cs} = 8 A/div), (T_e = 10 Nm/div).

on, the following observations are made for the proposed controller: (i) it eliminates overshoot in the rotor speed response with settling time of 60 ms, (ii) electromagnetic torque has smooth transient response i.e. overshoot of 60% and it reaches steady state with settling time of 50 ms, (iii) stator current has an overshoot of 60% with settling time of 50 ms. Further, the speed, electromagnetic torque and stator current response of proposed system is compared with PI-HCC, PI-DTC based PMSM drive during forward to reverse motoring operation and respective results are given in Table IV, V, VI. From the comparative studies significant observations for proposed controller are as follows: (i) the overshoot in the electromagnetic torque response has been reduced to 50% compared to PI-DTC and PI-HCC, (ii) the overshoot in stator current response is reduced to 30% compared to PI-DTC. Finally, the experimental studies confirms effectiveness of the proposed PDFF controller based speed regulator.

TABLE IV
TIME RESPONSE ANALYSIS OF THE SPEED UNDER DIFFERENT TEST CONDITIONS FOR PI-HCC, PI-DTC AND PDFF-DTC BASED PMSM DRIVE

Test condition	Time response	PI-HCC	PI-DTC	PDFF-DTC
Forward to reverse motoring operation	M_p (%)	5	5	0
	T_s (ms)	80	60	60
Reverse to forward braking operation	M_p (%)	140	130	56
	T_s (ms)	85	75	55

TABLE V
TIME RESPONSE ANALYSIS OF THE ELECTROMAGNETIC TORQUE UNDER DIFFERENT TEST CONDITIONS FOR PI-HCC, PI-DTC AND PDFF-DTC BASED PMSM DRIVE

Test condition	Time response	PI-HCC	PI-DTC	PDFF-DTC
Forward to reverse motoring operation	M_p (%)	125	120	60
	T_s (ms)	50	50	50
Reverse to forward braking operation	M_p (%)	105	135	20
	T_s (ms)	60	60	50

TABLE VI
TIME RESPONSE ANALYSIS OF THE STATOR CURRENT UNDER DIFFERENT TEST CONDITIONS FOR PI-HCC, PI-DTC AND PDFF-DTC BASED PMSM DRIVE

Test condition	Time response	PI-HCC	PI-DTC	PDFF-DTC
Forward to reverse motoring operation	M_p (%)	40	105	35
	T_s (ms)	50	50	50
Reverse to forward braking operation	M_p (%)	80	215	70
	T_s (ms)	60	60	50

V. CONCLUSION

This paper proposes PDFF based speed regulator which aims to reduce overshoot and oscillations in rotor speed, electromagnetic torque and stator current of direct torque controlled PMSM drive. The proposed PDFF-DTC is compared to the state of the art, which includes the PI-DTC and the PI-HCC. With the proposed controller during reverse to forward braking operation the overshoot in the rotor speed and electromagnetic torque response is reduced to 50% and 20%

respectively compared to other controllers. While in forward to reverse motoring operation the overshoot in the stator current response is reduced to 30% compared to other controllers. Thus, the experimental studies confirm effectiveness of the proposed PDFF controller based speed regulator for DTC-PMSM drive.

APPENDIX

PMSM parameters: $L_s = 24.3$ mH, $R_s = 1.15$ Ω , $\psi_m = 0.9426$ Wbturns, $B = 0.00029$ kg m²/s, $J = 0.0145$ kg m²

PMSM rating: 360 V, 3- ϕ , 1.5 kW, 50 Hz, 1500 Rpm, 4 pole.

DC machine rating: 2.2 kW, 1500 rpm, 4 pole, Armature: 200 V, 11 A; Field: 200 V, 0.9 A.

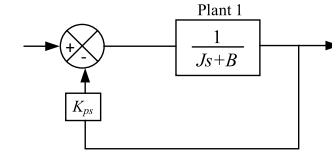
Power converter: 415V, 30A, 30 kHz switching frequency.

Resistive load: (0-165 Ω), 16 A.

A. Derivation for the Transfer Function of PDFF Speed Controller

By using block diagram reduction technique, Fig. 2 is simplified as:

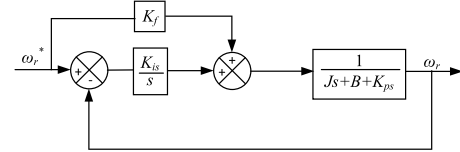
Loop 1:



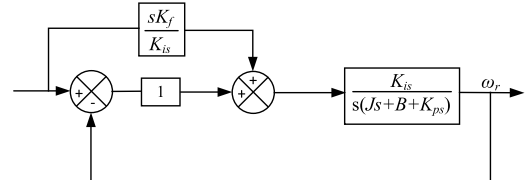
By solving loop 1:

$$\text{Loop1} = \frac{1}{Js + B + K_{ps}}$$

Substituting loop 1 term in the block diagram, then:



By simplifying above block diagram, then:



By solving above block diagram, the transfer function of PDFF speed controller is derived as:

$$\frac{\omega_r}{\omega_r^*} = \frac{K_f s + K_{is}}{Js^2 + (B + K_{ps})s + K_{is}}$$

B. Derivation for the Transfer Function of PI Speed Controller

By simplifying the forward path in Fig. 3.

$$\begin{aligned} G(s) &= \left(K_{ps} + \frac{K_{is}}{s} \right) \frac{1}{Js + B} \\ &= \frac{sK_{ps} + K_{is}}{s} \cdot \frac{1}{Js + B} \\ &= \frac{sK_{ps} + K_{is}}{Js^2 + sB} \end{aligned}$$

The closed loop transfer function of PI current controller is given below:

$$\begin{aligned}\frac{\omega_r}{\omega_r^*} &= \frac{G(s)}{1 + G(s)H(s)} \\ \frac{\omega_r}{\omega_r^*} &= \frac{\left(\frac{sK_{ps} + K_{is}}{Js^2 + sB} \right)}{1 + \frac{sK_{ps} + K_{is}}{Js^2 + sB}} \\ \frac{\omega_r}{\omega_r^*} &= \frac{sK_{ps} + K_{is}}{Js^2 + (B + K_{ps})s + K_{is}}\end{aligned}$$

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